

Volume III. Eighth Section. The group of the transformation equations and the equation of the fifth degree.

§ 79. Examination of the group $\mathfrak{L}_0$.

In the tenth section of Volume II we investigated the congruence group and its divisors quite generally. Here we carry out this investigation once more in a special form, insofar as it has reference to the transformation problem.

The substitutions contained in $\mathfrak{L}_0$,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \left(z, \frac{-c + az}{d - bz} \right), \quad (1)$$

in which

$$\Delta = ad - bc \quad (2)$$

is a quadratic residue of p , can, as already remarked above, be brought to the form

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} = \left(z, \frac{-\gamma + \alpha z}{\delta - \beta z} \right), \quad (3)$$

where

$$\alpha\delta - \beta\gamma \equiv 1 \pmod{p}. \quad (4)$$

For we need only, if by $\sqrt{\Delta} \pmod{p}$ we understand a whole number whose square is congruent to Δ modulo p , to put

$$a \equiv \alpha\sqrt{\Delta}, \quad b \equiv \beta\sqrt{\Delta}, \quad c \equiv \gamma\sqrt{\Delta}, \quad d \equiv \delta\sqrt{\Delta} \pmod{p}.$$

In the form (3) the signs of $\alpha, \beta, \gamma, \delta$ can still be changed simultaneously.

We ask for those elements z which remain unchanged by a substitution of the form (3). These are determined by the congruence

$$z \equiv \frac{-\gamma + \alpha z}{\delta - \beta z} \pmod{p},$$

or

$$\beta z^2 + (\alpha - \delta)z - \gamma \equiv 0 \pmod{p}, \quad (5)$$

a congruence which, unless it is identical, has at most two incongruent roots. To obtain these, under the supposition first that β is not divisible by p , we write (5) as

$$\left(\beta z + \frac{\alpha - \delta}{2} \right)^2 \equiv \left(\frac{\alpha - \delta}{2} \right)^2 + \beta\gamma \pmod{p},$$

or, with the help of (4),

$$\left(\beta z + \frac{\alpha - \delta}{2} \right)^2 \equiv \left(\frac{\alpha + \delta}{2} \right)^2 - 1 \pmod{p}. \quad (6)$$

Accordingly three different cases are to be distinguished:

$$\text{I.} \quad \left(\frac{\alpha + \delta}{2} \right)^2 - 1 \equiv 0 \pmod{p};$$

then the congruence (5) has one root; there is one and only one element z which remains unchanged.

$$\text{II.} \quad \left(\frac{\alpha + \delta}{2} \right)^2 - 1 \text{ is a quadratic residue of } p;$$

the congruence (5) has two different roots; there are two elements z which remain unchanged.

$$\text{III.} \quad \left(\frac{\alpha + \delta}{2}\right)^2 - 1 \text{ is a quadratic non-residue of } p;$$

the congruence (5) has no root at all, and all elements z are permuted.

If $\beta \equiv 0$, then (5) always has the one root $z = \infty$; if then $\delta = \alpha$ and $\gamma \not\equiv 0 \pmod{p}$, there is only this one. But in this case

$$\alpha\delta \equiv 1, \quad \alpha \equiv \delta, \quad \left(\frac{\alpha + \delta}{2}\right)^2 \equiv 1 \pmod{p},$$

and condition I is fulfilled. If however at the same time $\gamma \equiv 0 \pmod{p}$, the substitution is the identical one.

If $\beta \equiv 0$, but $\alpha - \delta \not\equiv 0 \pmod{p}$, then (5) has still a second root; since now $\alpha\delta \equiv 1 \pmod{p}$, we have

$$\left(\frac{\alpha + \delta}{2}\right)^2 - 1 \equiv \left(\frac{\alpha - \delta}{2}\right)^2 \pmod{p},$$

therefore a quadratic residue, and condition II is fulfilled. We summarize these propositions thus:

In case I one element, or all elements, remain unchanged; in case II two elements remain unchanged; and in case III all elements are changed.

If we repeat one and the same substitution

$$A = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \tag{7}$$

several times, the substitutions

$$A, A^2, A^3, \dots$$

arise, and in their series the identical substitution $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ must occur once. If n is the least positive number for which

$$A^n = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

then n is called the degree of A (Vol. II, § 2).

If for an arbitrary m we put

$$A^m = \begin{pmatrix} \alpha_m & \beta_m \\ \gamma_m & \delta_m \end{pmatrix},$$

then for the computation of the numbers $\alpha_m, \beta_m, \gamma_m, \delta_m$ we obtain the following system of recurrent formulae:

$$\begin{aligned} \alpha_{m+1} &= \alpha\alpha_m + \beta\gamma_m, & \beta_{m+1} &= \alpha\beta_m + \beta\delta_m, \\ \gamma_{m+1} &= \gamma\alpha_m + \delta\gamma_m, & \delta_{m+1} &= \gamma\beta_m + \delta\delta_m. \end{aligned} \tag{8}$$

The numbers $\alpha_m, \beta_m, \gamma_m, \delta_m$ are computed especially simply from these formulae in case I.

In this case, since the signs of $\alpha, \beta, \gamma, \delta$ may all be simultaneously reversed, we can assume

$$\alpha + \delta = 2, \quad \alpha\delta - \beta\gamma = 1 \pmod{p},$$

and from (8) obtain

$$\begin{aligned} \alpha_2 &\equiv -1 + 2\alpha, & \beta_2 &\equiv 2\beta, \\ \delta_2 &\equiv -1 + 2\delta, & \gamma_2 &\equiv 2\gamma, \\ \alpha_3 &\equiv -2 + 3\alpha, & \beta_3 &\equiv 3\beta, \\ \delta_3 &\equiv -2 + 3\delta, & \gamma_3 &\equiv 3\gamma, \end{aligned} \pmod{p},$$

from which, by the induction from m to $m + 1$, it follows that

$$\alpha_m \equiv -(m-1) + m\alpha, \quad \delta_m \equiv -(m-1) + m\delta, \quad \beta_m \equiv m\beta, \quad \gamma_m \equiv m\gamma \pmod{p}. \quad (9)$$

From this one sees that all the A^m again belong to case I, and that p is the degree of A .

The analogous consideration of the two other cases has been carried out by Serret, but appears unnecessary for our purpose.

On the other hand, we shall add here the theorem that the whole group $\mathfrak{L}_0$ can be composed from the following two special substitutions

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \quad (10)$$

The proof follows from § 50, if one observes that to each number system $\alpha, \beta, \gamma, \delta$ satisfying the condition $\alpha\delta - \beta\gamma = 1 \pmod{p}$, the number system $\alpha', \beta', \gamma', \delta'$ can be chosen so that

$$\begin{pmatrix} \alpha' & \beta' \\ \gamma' & \delta' \end{pmatrix} \equiv \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \pmod{p}, \quad \alpha'\delta' - \beta'\gamma' = 1.$$

Thus all substitutions of $\mathfrak{L}_0$ are contained among those considered in § 30. But we can also here easily display the composition of an arbitrary substitution in $\mathfrak{L}_0$ actually from A and B , and so give the proof independently of § 30.

For if first c is an arbitrary number not divisible by p , then actual calculation gives easily

$$A^c = \begin{pmatrix} 1 & 0 \\ c & 1 \end{pmatrix}, \quad C = \begin{pmatrix} c & 0 \\ 0 & c^{-1} \end{pmatrix} = A^{-c^{-1}} B A^{-c} B A^{-c^{-1}} B, \quad (11)$$

and then, if $\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$ is any substitution in $\mathfrak{L}_0$ and β is different from 0,

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} = \begin{pmatrix} \alpha & 0 \\ \delta\beta^{-1} - 1 & 1 \end{pmatrix} \begin{pmatrix} \beta & 0 \\ 0 & \beta^{-1} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \alpha\beta^{-1} & 1 \end{pmatrix}, \quad (12)$$

and if $\beta = 0$,

$$\begin{pmatrix} \alpha & 0 \\ \gamma & \alpha^{-1} \end{pmatrix} = \begin{pmatrix} \alpha & 0 \\ 0 & \alpha^{-1} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \gamma\alpha & 1 \end{pmatrix}. \quad (13)$$

It is to be concluded from this theorem that a group contained in $\mathfrak{L}_0$ which contains the two substitutions A, B must necessarily be identical with $\mathfrak{L}_0$.

§ 80. Normal divisors of the group $\mathfrak{L}_0$.

In order to learn the possible reductions of the transformation equation, it is above all necessary to investigate the divisors of the group $\mathfrak{L}_0$. We ask first about the existence of a normal divisor $\mathfrak{K}$ of $\mathfrak{L}_0$ (Vol. II, § 3).¹

Let

$$S = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \quad (1)$$

be any substitution in $\mathfrak{K}$ different from the identical substitution. Then, by the nature of a normal divisor, every substitution

$$U = TST^{-1} \quad (2)$$

¹According to Galois: "proper divisors".

is also contained in $\mathfrak{K}$, if

$$T = \begin{pmatrix} \alpha' & \beta' \\ \gamma' & \delta' \end{pmatrix} \quad (3)$$

is an arbitrary substitution in $\mathfrak{L}_0$.

We set ourselves the problem of determining T and ξ so that

$$U = \begin{pmatrix} 0 & 1 \\ -1 & \xi \end{pmatrix} \quad (4)$$

shall hold. This condition can also be written

$$\begin{pmatrix} \alpha' & \beta' \\ \gamma' & \delta' \end{pmatrix} \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & \xi \end{pmatrix} \begin{pmatrix} \alpha' & \beta' \\ \gamma' & \delta' \end{pmatrix},$$

and leads to the congruences

$$\begin{aligned} 1. \quad & \alpha'\alpha + \beta'\gamma \equiv \gamma', \\ 2. \quad & \alpha'\beta + \beta'\delta \equiv \delta', \\ 3. \quad & \gamma'\alpha + \delta'\gamma \equiv -\alpha' + \gamma'\xi, \\ 4. \quad & \gamma'\beta + \delta'\delta \equiv -\beta' + \delta'\xi, \end{aligned} \pmod{p}, \quad (5)$$

and from 1 and 2 there also follows

$$\alpha' \equiv \gamma'\delta - \delta'\gamma, \quad \beta' \equiv -\gamma'\beta + \delta'\alpha \pmod{p}. \quad (6)$$

If these values are inserted in the third and fourth congruences of (5), there follows

$$\gamma'(\alpha + \delta - \xi) \equiv 0, \quad \delta'(\alpha + \delta - \xi) \equiv 0 \pmod{p},$$

whence, since γ', δ' cannot both be divisible by p , it follows that

$$\xi \equiv \alpha + \delta \pmod{p}. \quad (7)$$

If ξ is so determined, then in (5) the congruences 3 and 4 follow from 1 and 2. If one now inserts α', β' from 1 and 2 in

$$\alpha'\delta' - \beta'\gamma' \equiv 1 \pmod{p}, \quad (8)$$

one obtains

$$\alpha'^2\beta + \alpha'\beta'(\delta - \alpha) - \beta'^2\gamma \equiv 1 \pmod{p}. \quad (9)$$

If α', β' are determined from this, all demands stated in (5), and hence also in (2), (4), are satisfied.

It remains therefore only to prove that the congruence (9) is always soluble.

This possibility is evident if β or $-\gamma$ is a quadratic residue of p ; for then

$$\alpha' \equiv \sqrt{\beta^{-1}}, \quad \beta' = 0,$$

or

$$\alpha' = 0, \quad \beta' \equiv \sqrt{-\gamma^{-1}} \pmod{p}$$

satisfies the demand.

It is further to be distinguished whether the substitution S belongs to class I, II, or III of the preceding paragraph.

First, if

$$\text{I.} \quad \left(\frac{\alpha + \delta}{2} \right)^2 \equiv 1 \pmod{p},$$

then, if S^m is put in place of S , by § 79, (9), β passes into $m\beta$ and γ into $m\gamma$; and m can always be so determined that $m\beta$ or $-m\gamma$ becomes a quadratic residue of p . This also includes the case in which $\beta = 0$ or $\gamma = 0$ and $\alpha \equiv \delta \pmod{p}$.

If, on the other hand, $\beta = 0$ and $\alpha - \delta$ is not divisible by p , then one can choose β' arbitrarily in (9) and then determine α' from a congruence of the first degree.

If β is not divisible by p , put (9) in the form

$$\left[\alpha' \beta + \beta' \frac{\delta - \alpha}{2} \right]^2 \equiv \beta + \beta'^2 \left[\left(\frac{\alpha + \delta}{2} \right)^2 - 1 \right] \pmod{p}. \quad (10)$$

If now

$$\text{II.} \quad \left(\frac{\alpha + \delta}{2} \right)^2 - 1 \quad \text{is a quadratic residue of } p,$$

then β' is determined from the congruence

$$\beta'^2 \left[\left(\frac{\alpha + \delta}{2} \right)^2 - 1 \right] \equiv \left(\frac{\beta - 1}{2} \right)^2,$$

whereby (10) passes into

$$\alpha' \beta + \beta' \frac{\delta - \alpha}{2} \equiv \pm \frac{\beta + 1}{2} \pmod{p},$$

and from this α' can be determined, whichever sign is chosen.

If

$$\text{III.} \quad \left(\frac{\alpha + \delta}{2} \right)^2 - 1 \quad \text{is a quadratic non-residue of } p,$$

and at the same time β is a quadratic non-residue, then a non-residue ν can always be so determined that $\beta + \nu$ becomes a quadratic residue; for if ν in $\beta + \nu$ is allowed to run through the series of non-residues, not all non-residues can arise, since among them β too would have to occur.

Then β' can be so determined that

$$\beta'^2 \left[\left(\frac{\alpha + \delta}{2} \right)^2 - 1 \right] \equiv \nu \pmod{p},$$

and then α' from

$$\alpha' \beta + \beta' \frac{\delta - \alpha}{2} \equiv \pm \sqrt{\beta + \nu} \pmod{p}.$$

From this follows:

In a normal divisor $\mathfrak{K}$ of the group $\mathfrak{L}_0$ there must certainly occur a substitution U of the form

$$\begin{pmatrix} 0 & 1 \\ -1 & \xi \end{pmatrix}. \quad (11)$$

Let first ξ be different from zero modulo p .

By the concept of the normal divisor, $\mathfrak{K}$ also contains the substitution

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & \xi \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} \xi & 1 \\ -1 & 0 \end{pmatrix}, \quad (12)$$

and consequently also

$$\begin{pmatrix} 0 & 1 \\ -1 & \xi \end{pmatrix} \begin{pmatrix} \xi & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ \xi^2 - 1 & -1 \end{pmatrix} = A^{\xi^2 - 1} \quad [\S 79, (11)], \quad (13)$$

hence also A and all its powers. Therefore $\mathfrak{K}$ also contains

$$\begin{pmatrix} 1 & 0 \\ \eta & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & \xi \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & \xi + \eta \end{pmatrix} \quad (14)$$

for arbitrary η , that is, the substitution (11) for every arbitrary ξ , and hence also

$$B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Accordingly, by the theorem of the preceding paragraph, the group $\mathfrak{K}$ is identical with $\mathfrak{L}_0$.

There remains still the possibility to discuss in which, in (11), $\xi = 0$.

In this case the group $\mathfrak{K}$ contains the substitution

$$B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

and consequently, for an arbitrary α not divisible by p , also

$$V = \begin{pmatrix} 0 & \alpha \\ -\alpha^{-1} & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -\alpha \\ \alpha^{-1} & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} \alpha^2 & 0 \\ 0 & \alpha^{-2} \end{pmatrix}.$$

From this one derives further, as contained in $\mathfrak{K}$,

$$W = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} \alpha^{-2} & 0 \\ 0 & \alpha^2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \alpha^2 & 0 \\ 0 & \alpha^{-2} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ \alpha^4 - 1 & 1 \end{pmatrix}.$$

If now p is greater than 5, then α can be so taken that $\alpha^4 - 1$ is not divisible by p ; then $\mathfrak{K}$ also contains the substitution A and its powers, and hence $\mathfrak{K}$ is identical with $\mathfrak{L}_0$.

If $p = 5$, then $\alpha^4 - 1$ is always divisible by p , so this conclusion is not applicable. But in this case it follows from the concept of the normal divisor that the following is contained in $\mathfrak{K}$:

$$\begin{aligned} & \begin{pmatrix} 1 & 1 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} -2 & -1 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} -2 & 0 \\ 0 & 2 \end{pmatrix}, \\ & \begin{pmatrix} -2 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}, \end{aligned}$$

that is, A^2 , and hence all powers of A . From this one concludes as above that $\mathfrak{K}$ is identical with $\mathfrak{L}_0$. We have therefore the theorem:

If $p > 3$, the group $\mathfrak{L}_0$ has no normal divisor.

In the case $p = 3$, $\mathfrak{K}$ likewise contains

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

and therefore also

$$\begin{pmatrix} \pm 1 & 0 \\ \pm 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \mp 1 & 1 \end{pmatrix} = \begin{pmatrix} \mp 1 & 1 \\ 1 & \pm 1 \end{pmatrix},$$

and in fact the four substitutions

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (15)$$

form a normal divisor of $\mathfrak{L}_0$ of index 3.

In the case $p = 3$, the group $\mathfrak{L}_0$ is isomorphic with the alternating group of interchanges of four elements. This is the group of an arbitrary equation of the fourth degree when the square root of the discriminant is adjoined to the domain of rationality. The divisor (15) of this group of index 3 then supplies the cubic resolvent of the biquadratic equation known in algebra.

A function belonging to the group (15) of the roots v_∞, v_0, v_1, v_2 of the transformation equation for the third transformation degree is, for example,

$$v_\infty v_0 + v_1 v_2$$

or

$$(v_\infty - v_0)(v_1 - v_2).$$

The difference between these two functions is that the first remains unchanged by the substitution of determinant -1

$$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix},$$

whereas the second changes its sign under it. The first will therefore lead to an equation of the third degree with rational coefficients, while in the cubic equation for the second $\sqrt{-3}$ still occurs (§ 78).